Experiment - 1

Addition

Aim:- To write an R program to perform addition of two numbers.

Program:-

```
num1 ← 10
num2 ← 20
x ← num1
y ← num2
z ← x+y
Print (z)
```

output

```
Print(z)
=30.
```

Subtraction - Experiment - 2

Aim:- To write an R program to perform subtraction of two numbers.

Program:-

```
num1 ← 10
num2 ← 20
x ← num1
y ← num2
z ← x-y
Print (z)
```

output

```
Print(z)
=10.
```

Experiment -3

Multiplication

Aim:- To write an R program to perform Multiplication of two numbers.

Program:-

```
num1 ← 10
num2 ← 20
x ← num1
y ← num2
z ← x*y
print(z)
```

Output

```
print(z)
= 200
```

Experiment -4

Division

Aim:- To write an R program to perform Division of two numbers.

Program:-

```
num1 ← 10
num2 ← 20
x ← num1
y ← num2
z ← x/y
print(z)
```

Output

```
print(z)
= 2
```

Experiment-5

Odd or even

Aim:- To write an R Program to check whether a number is odd or even.

Program:-

```
num ← 10
x ← num
if (x%.%.2 == 0) {
    print ("even")
} else {
    print ("odd")
}.
```

Output:-

even.

Experiment-6

Mean, Median, Mode

Aim:- To write an R program to find the Mean, Median, Mode of a dataset.

Program:-

```
x ← c(10, 20, 30, 40, 50)
print(mean(x))
print(median(x))
print(mode(x))
```

Output

```
Mean = 30
median = 30
Mode = nomeric.
```

Experiment - 7

Summary

Aim:- To write an R program to display the summary of a dataset.

Program:-

```
x ← c(10, 20, 30, 40, 50)
summary(x)
```

Output:-

```
Min  1st Q  Median  Mean  3rd Q  M
 10   20     30      30    40     50
```

Experiment - 8

Greatest Among three numbers

Aim:- To write an R program to find the greatest among three numbers.

Program:-

```
a ← 10
b ← 20
c ← 30

x ← a
y ← b
z ← c
if (x>y && x>z){
  Print(x)
} else if (y>z) {
  Print(y)
} else {
  Print(z)
```

Output

```
Print(z)
  30
```

Experiment-9

IQR

Aim:- To write an R Program to find the interquartile Range (IQR)

Program:-

```
x ← c(10,20,30,40,50)
Print (IQR (x))
```

Output:-

```
Print(IQR(x)) =20
```

Experiment-10

Quantile

Aim:- To write an R program to find the quantiles of a dataset.

Program:-

```
x ← c (10,20,30,40,50)
print (quantile (x))
```

output

0%.	25%.	50%.	75%.	100%.
10	20	30	40	50.

Experiment-11

Mid Range

Aim:- To write an R program to find the Mid Range of a dataset.

Program:-

```
x ← c(10, 20, 30, 40, 50)
a ← max(x)
b ← min(x)
z ← (a+b)/2
print(z)
```

Output:

```
print(z) = 30
```

Experiment-12

Z-Score Normalization

Aim:- TO write an R program for z-score Normalization.

Program:-

```
x ← c(10, 20, 30, 40, 50)
a ← mean(x)
b ← sd(x)
z ← (x-a)/b
print(z)
```

output

```
-1.2649111  -0.6324555
0.0000000   0.6324555
1.2649111
```

Experiment-13

Min, Max, Mean and Min-Max Normalization

Aim:- To write an R-program to find Maximum, Minimum, Mean, Min-Max

Program:-

```
x ← c(10,20,30,40,50)
a ← min(x)
b ← max(x)
c ← mean(x)
z ← (x-a)/(b-a)
print(a)
print(b)
print(c)
print(z)
```

output

0.00 0.25 0.50 0.7
1.00

Experiment-14

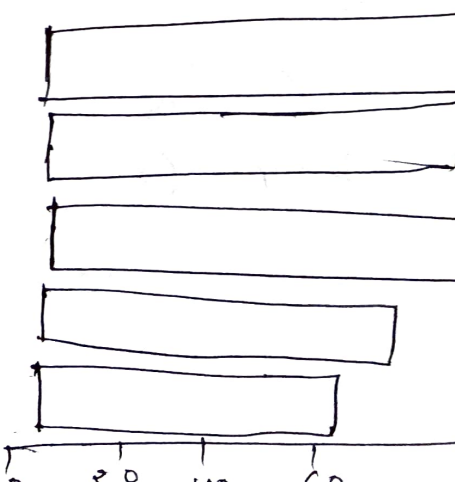
Barplot and Horizental Barplot

Aim:- To write an R program to draw a Barplot and Horizental Barplot.

Program:-

```
a ← c(55, 67, 89, 80, 90)
barplot(a)
barplot(a, horiz=TRUE)
```

output



Experiment-15

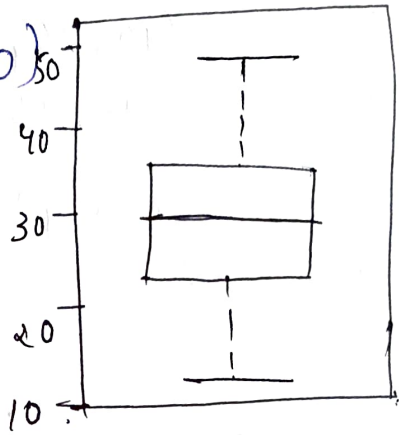
Box plot

Aim:- To write an R program to draw a Box plot.

program:-

```
x ← c(10,20,30,40,50)
boxplot(x).
```

output



Experiment-16

Histogram

Aim:- To write an R program to draw a Histogram.

program:-

```
x ← c(10,20,30,40,50)
hist(x)
```

output:-

Experiment -17

Correlation

Aim:- To write an R program to perform Correlation Analysis.

Program:-

```
x ← c(10, 20, 30, 40, 50)
y ← c(15, 25, 35, 45, 55)
z ← cor(x, y)
print(z)
```

output:-

```
print(z) = 1
```

Experiment -18

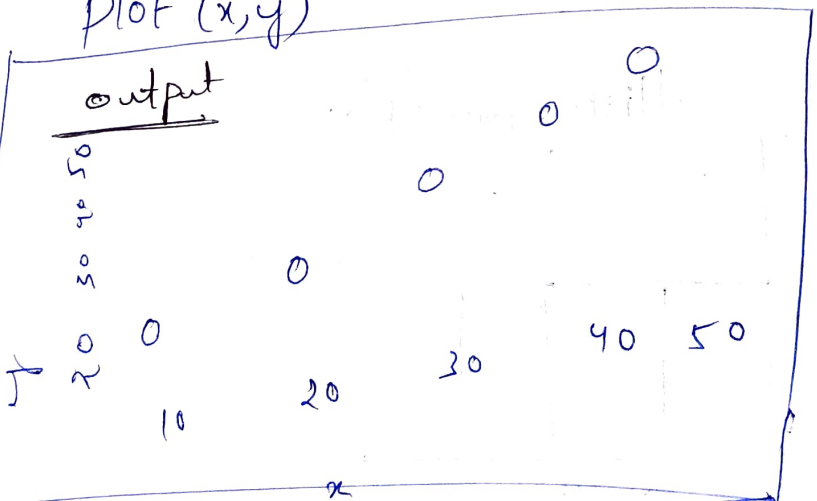
Scatterplot

Aim:- To write an R program to draw a scatter plot.

Program:-

```
x ← c(10,20,30,40,50)
y ← c(15,25,35,45,55)
plot(x,y)
```

output



Experiment-19

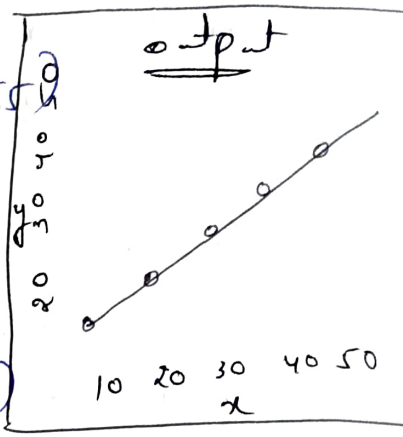
Linear Regression

Aim:- To write an R program to perform Linear Regression.

Program:-

```
x ← c(10, 20, 30, 40, 50)
y ← c(15, 25, 35, 45, 55)
model ← lm(y ~ x)
print(summary(model))
plot(x,y)
abline(model, col = "red")
```

output



Experiment-20

Multiple Regression

Aim:- To write an R program to perform Multiple Regression.

Program:-

```
data ← data.frame(
  Age = c(20,25,30,35,40)
BP = c(120, 125, 130, 135, 140)
Glucose = c(80, 90, 100, 110, 120)
x ← data
model ← lm(Age ~ BP + Glucose, data = x)
print(summary(model))
```

output

```
Call:
lm(formula = Age ~ BP + Glucose,
data = data)
```

Weka

Experiment - 21.

Apriori Algorithm.

Aim:- To find frequent itemsets using Apriori algorithm.

Algorithm:-

1. Input transaction dataset
2. Generate L1 (frequent-1-itemsets)
3. Generate L2
4. Generate L3
5. Stop when no more itemsets.
6. Generate rules.

Notepad code:-

```
@ relation apriori.
@ attribute Rice {yes, no}
@ attribute sugar {yes,no}
@ attribute milk{yes,no}
@ attribute Tea {yes,no}
@ attribute Bread {yes,no}
@ attribute Butta {yes,no}
@ data
 yes, yes, yes, yes, no,no, no
 yes, no, yes, yes, yes, no,no
  no, no, yes, no, yes, yes, no
  yes, yes, no, yes, yes, no.
```

Output:-

1. Bread=no 6 => Rice=yes6
2. Rice = yes6 => Bread=no 5
3. Rice= yes6 => Butta=no 6
4. Sugar=yes6 => Butta=no 6
5. Bread=no6 => Butta=no6

Result:-

Frequent pattern Exist among Rice, sugar, Milk.

Experiment-22

Aim:- To generate frequent itemsets using Apriori

Algorithm:-

1. Input dataset
2. Apply Apriori
3. Apply FP-Growth
4. Compare results.

Notepad code:-

```
@ relation apriori 2
@ attribute B {yes,no}
@ attribute R {yes,no}
@ attribute E {yes,no}
@ attribute A {yes,no}
@ attribute D {yes,no}
@ attribute M {yes,no}
@ attribute U {yes,no}
@ attribute T {yes,no}
@ data
```

Output:- Apriori

1. I=no 5 ==> B=yes 5
2. I=no 5 ==> B=yes 5
3. K=no 5 ==> B=yes 5
10. I=no L=no 5 ==> B=yes 5

Fp-Growth

1. (E=no):5 ==>
(K=no):5
(K=no):5 ==> (L=no):5
(I=no):5 ==> (I=no):5
(I=no):5 ==> (L=no, K=no)

Result:-

Fp growth gives same result but faster.

Experiment 2

Aim:-
To classify data using Decision Tree and logistic Regression.

Algorithm:-
1. Load dataset
2. Preprocess data
3. Apply J48
4. Apply logistic
5. Compare accuracy

Notepad code:-

@relation bank
@ attribute age numeric
@ attribute balance numeric
@ attribute loan {yes, no}
@ attribute default {yes, no}
@data
25, 1000, no, no
30, 1500, no, no
35, 2000, yes, yes
40, 2500, no, no
45, 3000, yes, yes
50, 3500, no, no
55, 4000, yes, yes
60, 4500, no, no

Output:-
Decision Tree (J48)
Correctly classified Instance = 5 (62.5%)
Incorrectly classified Inst = 3
Kappa stastic = 0
Total Instance = 8

Logistic Regr
= 8 (10
= 0
= 1
= .8

Result :- Logistic Regression perform

Experiment-24

Aim:- To generate frequents itemsets and rules from book dataset (Books ARFF + Association Rule).

Algorithm:-

1. Create ARFF
2. Apply Apriori
3. Apply FP-growth
4. Compare.

Notepad code:-

```
@relation books
@attribute Datamining {yes,no}
@attribute DBMS {yes,no}
@attribute AI {yes,no}
@attribute Os {yes,no}
@data
yes  yes  yes  no
no   yes  no   yes
yes  no   yes  no
no   yes  yes  no
yes  yes  no   yes
no   no   yes  yes.
```

Output Apriori : $L_1 = 8, L_2 = 13, L_3 = 3$

Rules:

Os => AI
AI => DBMS
DBMS => AI

Result:- Apriori and FP-growth produce similar rule.

Experiment 25

Aim:- To compare Decision Tree and Rule based classification.

Algorithm:-

1. load dataset
2. Apply J48
3. Apply JRiD / PART
4. Compare accuracy.

Notepad code:-

```
@relation credit.
@attributes age numeric
@attributes income numeric
@attributes student {yes, no}
@attributes rating {fair, excellent}
@attributes class {good, bad}
@data.
25, 1000, no, fair, good
30, 4000, no, Excellent, good
35, 5000, Yes, fair, bad
40, 6000, no, fair, good
45, 7000, Yes, Excellent, bad
50, 8000, no, Excellent, good.
55, 9000, Yes, fair, bad
60, 10000, no, fair, good.
```

Output:-

	Decision Tree (J48)	Rule-based (JRIP)	Rule based (PA
Accuracy	62.5%	100%	62.5
Correctly class	5	8	3
Incorrect	3	0	

Result:-

JRip classifier performs better than J48, PART.

Experiment - 9b

Aim:- TO create ARFF dataset and compare Navi Bayes and decision Tree Using accuracy F1-score

Algorithm steps:-

1. convert dataset into ARFF format.
2. Load dataset in weka
3. Apply Naive Bayes J48 classifier.
4. compare accuracy & F1-score.
5. Identify best classifier.

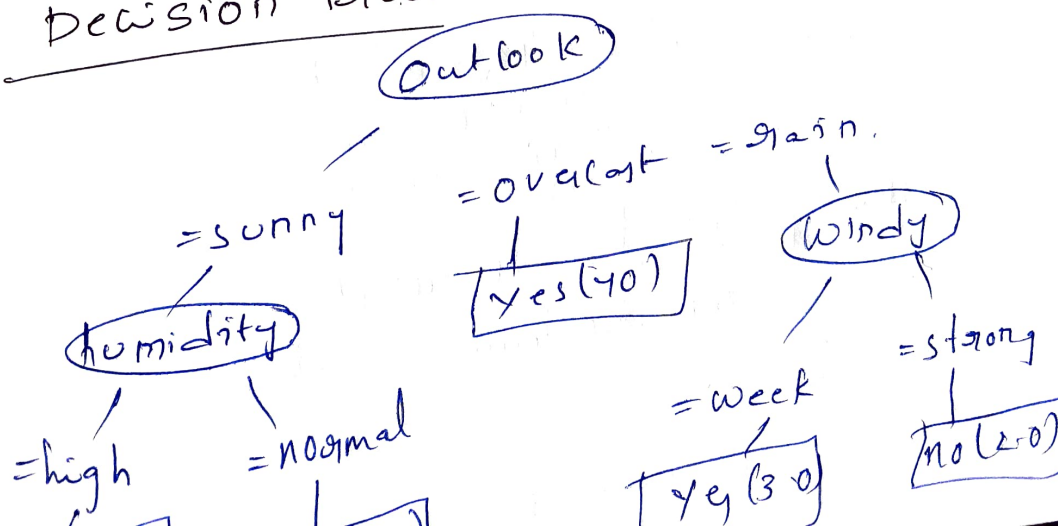
output:-

Decision Tree (J48)
correctly classified Instance = 8
Accuracy = 57.14%.
Incorrectly classified Instance = 6
F1 - Measures = 0.508

Confusion Matrix:-

a	b	
5	4	a = yes
3	2	b = no

Decision Tree:-



Naive Bayes output

Correctly classified instance = 7

Accuracy = 50%

Incorrect classified instance = 7

F_1 - Measures = 0.539

Confusion matrix

a	b	
7	2	a = yes
4	1	b = no

Result:- Decision Tree classifier Performed better than Naive Bayes because it achie higher accuracy and F1-S

Experiment - 27

Aim:- To perform K-means clustering and Visualize employee data in weka

Algorithm steps:

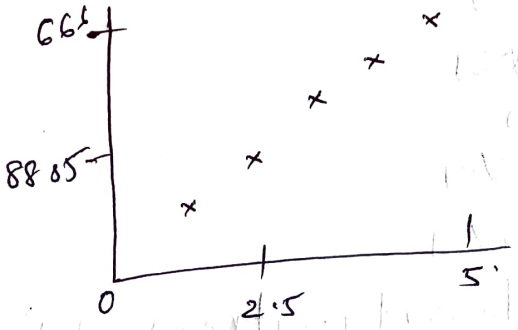
1. create csv dataset
2. load dataset into Weka
3. Apply simple kmeans clustering
4. set number of clusters.
5. Visualize clusters.

Notepad code:-

Employee ID, Gender, Age, salary, credit

111, Male, 28, 150000, 39
222, Male, 25, 150000, 27
333, Female, 26, 160000, 42
444, Female, 25, 160000, 40
555, Female, 30, 120000, 64
666, Male, 29, 20000, 72

Output:- cluster visualize.



Result:- Employees are grouped Successfully based on Salary and credit score.

Experiment -28

Aim:- To predict diabets using Dec. Tree and support Vector Machin classifies in weka

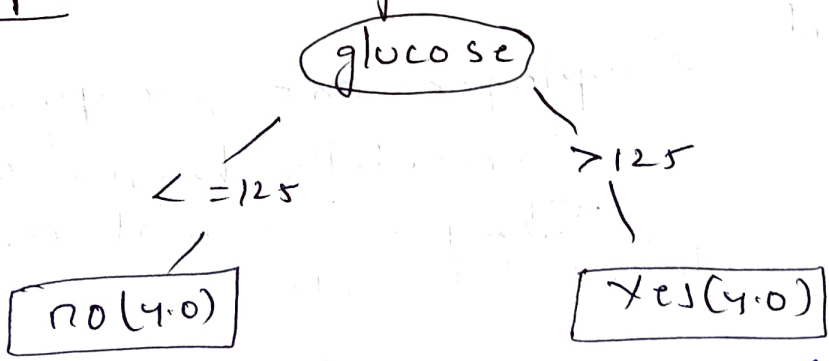
Algorithm Steps:-

1. load diabety dataset
2. Apply J48 Decission tree.
3. Apply SMO (SVM)
4. compare Accuracy and Fiscore

Notepad code:-

```
@ relation diabetes
@ attribute glucose numaric
@ attribute bmi numeric
@ attribute age numeric
@ attribute diabety {yes,no}
@data
120,25,30,no
140,30,35,yey
130,28,40,yey
110,22,25,no
150,35,50,yey
160,38,55,yey
```

Output:- Classify → trees → J48 (Decission Tree)



Result:- SVM classify performs better

Experiment-2g

Aim:- To identify the most Significant attribute using infogain evaluator.

Algorithm steps:-

1. Create ARFF dataset
2. Load dataset
3. Open Select Attributes tab
4. Choose info gain AttributeEval
5. Apply Ranker Search method.
6. Rank attributes.

Notepad code:-

```
@relation studentgrades
@attribute Marks numeric
@attribute Attendance numeric
@attribute Result {Pass,Fail}
@data
85,90,Pass
78,85,Pass
92,45,Pass
35,40,Fail
88,92,Pass
30,35,Fail
25,30,Fail
```

Output:- Ranked attributes

0.971 2 attendance

0.921 1 Marks

Selected attributes: 2, 1, 2

Result:- Marks attribut has the better information gain -

Experiment-30

Aim:- To perform k-means clustering and visualize clusters in weka

Algorithm steps:-

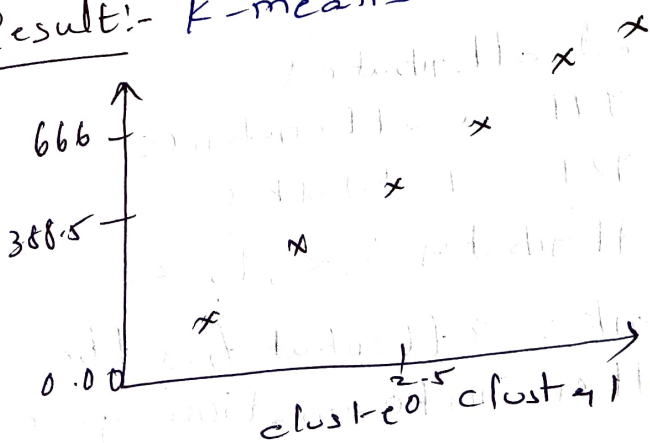
1. Create csv dataset
2. Load dataset into weka
3. Apply simple Means.
4. Set cluster size
5. Visualize output.

Notepad code:-

```
employee ID, Gendy, Age, Salary, Credit
111, Male, 28, 150000, 39
222, Male, 25, 150000, 27
333, Female, 26, 160006, 42
444, Female, 25, 160000, 40
555, Female, 30, 170000, 64
660, Male, 21, 200000, 72
```

Output:- 3 clusters formed
High Salary and low Salary groups identified.

Result:- k-means successfully



Experiment-3

Aim:- To generate rules from a Decision Tree and compare with Rule-Based induction.

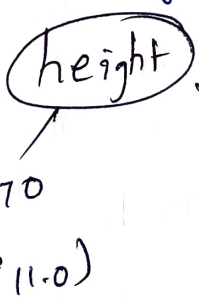
Algorithm steps.

1. Create ARFF dataset
2. Load into Weka
3. Apply J48
4. Apply JRip/PART
5. Compare accuracy and confusion Matrix.

Notepad code

```
@relation gender
@attribute height numeric
@attribute weight numeric
@attribute gender {male, female}
@data
185, 85, male
170, 60, female
182, 90, male
160, 55, female
190, 100, male
165, 58, Female
175, 70, male
158, 50, female.
```

Output:- classify → J48 → start (Visualize tree)



Result: JRip generated accurate rule and performed better than J48